CSE 5095: Temporal Matrix Factorization for High-dimensional Time Series Prediction

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1 Introduction

In recent years, time series prediction plays a critical role for a wide range of applications such as demanding prediction and traffic prediction. With the advent of big data, these problems are becoming increasingly high-dimensional, which requires to deal with a very large number(n) of time series with implicit or explicit inter-dependency. Not only is the scale of such problems huge, but they might involve missing values caused by human errors or sensor fault.

To cope with the high-dimension and sparse, a natural way is using matrix factorization to model the time series. [4, 6, 8] The time series matrix is decomposed into two small latent factor matrix with dimension K, where K is much smaller than n. Those traditional matrix factorization based approaches efficiently solved above problems with linearly increasing time complexity in n. However, those approaches fail to consider the temporal dependency among the latent factors.

Most recently, Hsiang-Fu et al. [7] proposed a temporal regularized matrix factorization(TRMF) model designing a temporal regularizer to encourage specific temporal structures among the factors, which shows superiority than other competitive methods. Although this model improved the accuracy than other matrix factorization based methods, the regularized approach is not expressive for long term dependency and is less flexible to extend on other dependency like spatial dependency.

In this paper, we propose a temporal non-regularized matrix factorization(TNMF) framework for high-dimensional time series prediction. Compared with other matrix factorization approaches, there are three contributions of this framework:

1. We present a new matrix factorization structure to capture the temporal dependency without any regularizer. The new approach directly embeds the temporal dependency into the latent factors by adding multiple delay factors, which is more expressive and flexible compared with regularized approaches.

2. We attempt to introduce other machine learning technologies and graph algorithms and integrate other additional information such as the explicit inter-dependency of one dimensional time series. In order to rectify the information loss caused by the matrix factorization and improve the accuracy.

3. A series of empirical analysis explains the rationality of the structure of our proposed TNMF model. The experiment on real-word dataset shows out framework can achieve better results than other similar methods.

2 Framework

Given a high-dimensional and sparse time series as $Y_{i,t}$ where i indicates the i-th one-dimensional time series and t indicates the t-th time point. There are four status $\{1, 2, 3, 4\}$ of $Y_{i,t}$ that represent four different states. Besides, the pre-defined internal relationship among all one-dimensional time series is described as graph $G = (V, E)$. This paper intends to predict for the future status of $Y_{i,\hat{t}}$ based on both Y and G.

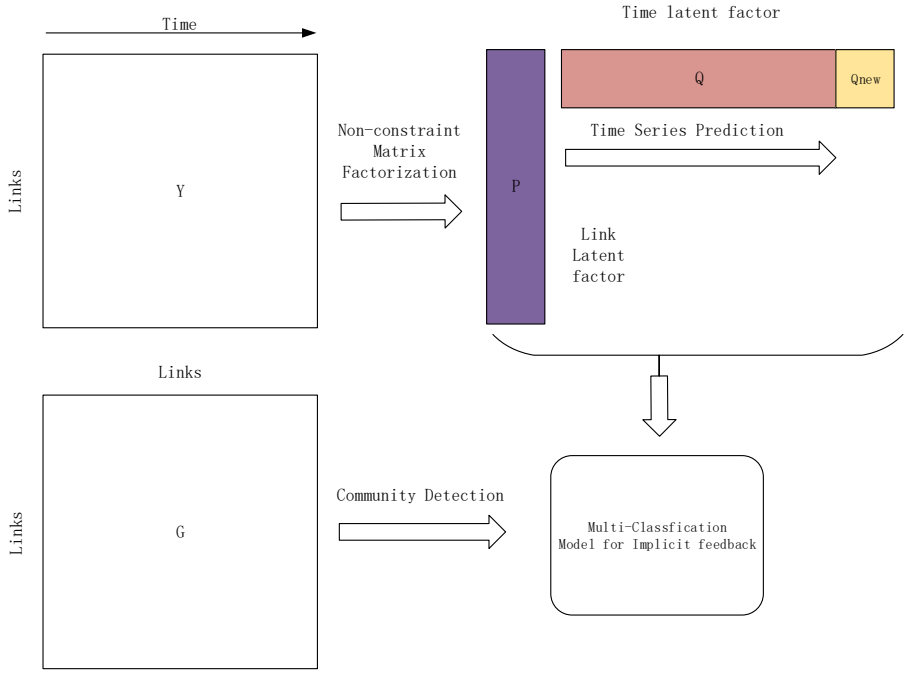


Figure 1: The main flow of TNMF framework for time series prediction.

Fig.1 illuminates the overall structure of our framework. First, we proposed a non-constraint matrix factorization to generate the time latent factors for each time point and link latent factors for each link. Then, we adopt a time series prediction method to generate the future time latent factor. Next, community detection is introduced to extract the internal relationship between time series. Finally, the prediction is made by a multi-classification model to overcome the implicit feedback problem.

3 Temporal Matrix Factorization

In this section, the motivation is using temporal matrix factorization converts the high-dimensional and sparse time series into the low-dimensional and dense latent factors, in order to tackle the missing value and speed up.

3.1 Temporal regularized matrix factorization

Matrix factorization is naturally a useful method to deal with the sparse and high-dimensional. Hsiang-Fu et al. proposed a temporal regularized matrix factorization(TRMF) framework for high-dimensional time series prediction by introducing the model based regularizer. In the specified autoregressive temporal sense, the problem is defined as:

$$\min_{P,Q,\mathcal{W}} \sum_{(i,t)\in\Omega} \left(Y_{it} - P_i^\top Q_t\right)^2 + \lambda_p \mathcal{R}_p(P) + \sum_{r=1}^{n} \lambda_q \mathcal{T}_{\mathrm{AR}}\left(\overline{Q}_r \mid \mathcal{L}, \overline{\boldsymbol{w}}_r, \eta\right) + \lambda_w \mathcal{R}_w(\mathcal{W}) \tag{1}$$

where P_i is the link latent factor of i-th link and Q_t indicates the time latent factor of t-th time point. W represents the weight of AR model and $\overline{Q}_r$ is the prediction of r-th time point from AR model. $\mathcal{L}$ is the leg set describes the certain previous time lag used in the autocorrelation. Fig.**??** shows the basic idea of TRMF framework.

The author also proposed the correlative algorithm to optimize P, Q, and W with total time complexity of $O\left(|\Omega|k^2\right) + O\left(|\mathcal{L}|Tk^2\right) + O\left(|\mathcal{L}|^3 + T|\mathcal{L}|^2\right)$

3.2 Temporal non-regularized matrix factorization

To incorporate the temporal dependency directly into the matrix factorization model, a temporal non-regularized model is proposed as follow:

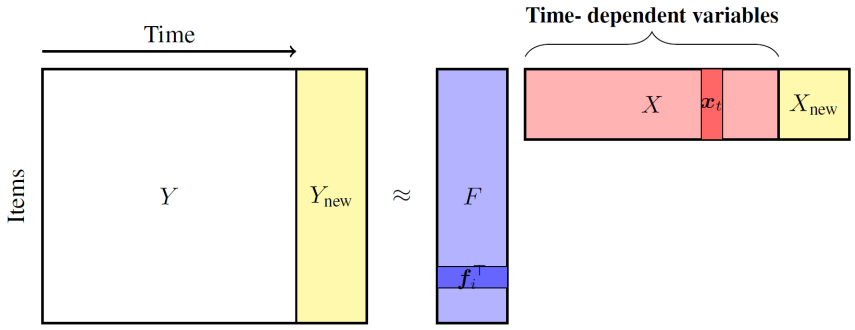


Figure 2: The basic idea of TRMF framework.

$$\min_{P,Q} \sum_{(i,t)\in\Omega} \left(Y_{it} - P_i^\top \sum_{r\in\mathcal{L}} Q_t^{(r)} \right)^2 + \lambda_p \mathcal{R}_p(P) + \lambda_q \mathcal{R}_q(Q) \tag{2}$$

where $Q_t^{(r)}$ is the r-th delay latent factor of time point t, which describes the contribution from the $t-r$ time point to t time point. The idea of TNMF is shown in Fig.3

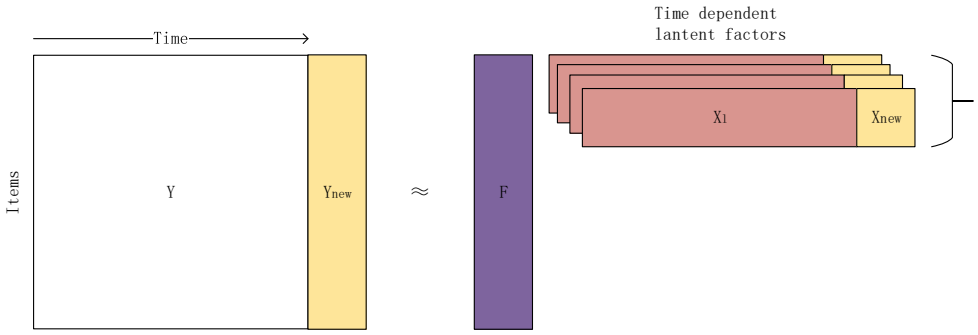


Figure 3: The basic idea of our TNMF model.

By minimizing (2), the parameters P and Q can be estimated iteratively as follows:

- Updates for P. When Q is fixed, the updating for P can be solved by least squire optimization, which costs $O(|\Omega|k^2)$.

- Updates for Q. When P is fixed, the updating for Q is solved iteratively by updating from $Q^{(0)}$ to $Q^{(|\mathcal{L}|)}$. The time complexity is $O(|\mathcal{L}| \times |\Omega| k^2)$

4 Time Series Prediction

Unlike TRMF, the main drawback of this work is that it fails to predict for the future based on its parameters. Hence, two prediction methods are adopted to predict the future time latent factors.

4.1 Cyclical factor prediction

Cyclical factors prediction is intuitive but useful method to capture the periodicity in time series. The basic idea is decomposing the time series as the product of average and cyclical factors. Then the future is forecasted by the moving average times the cyclical factors. The major disadvantage of this method is easy to implement and able to predict for time series with any length, but of particular concern is that the periodicity must be observed and assigned by hands first. Moreover, the accuracy of this model is limited compared with machine learning based methods due to its simpleness.

4.2 Seq-to-seq prediction

Seq-to-seq [5] model offers an effective way of long time series prediction. Particularly in this framework, the chosen seq2seq model has two main parts: encoder and decoder. The encoder is composed by GRU cells and the decoder is GRU block cells as Fig.4.

Seq-to-seq can be thought as an end to end and well-studied deep learning based time series model, which is much more flexible and expressive than statistical models. Another advantage is that this model is all the hype nowadays and able to capture the non-linear temporal relations compared with models like cyclical factor prediction or AR.

5 Inner Dependency Representation

In addition to the temporal dependency, another dependency is the relationship among all single time series. Those relationship can be divided as two types, one is explicit relationship that pre-defined by certain problem, for

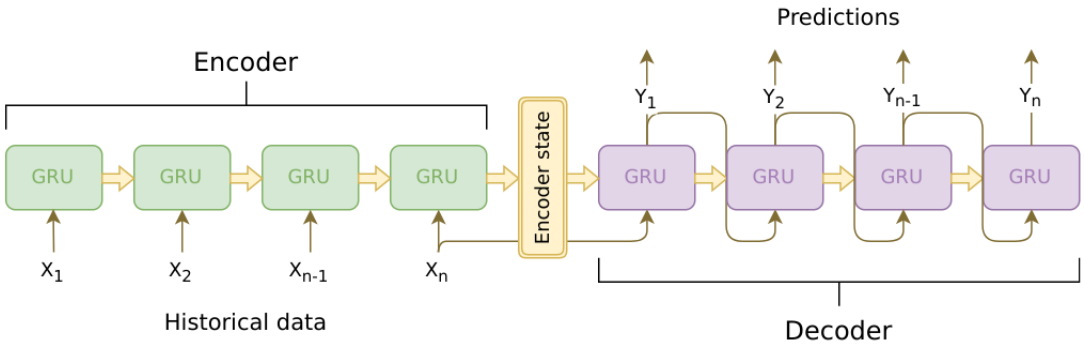


Figure 4: The structure of Seq-to-seq model.

example, topological network of road links in traffic dataset or the molecule structure of protein. Another one is the implicit relationship hidden in the high-dimensional time series, for example, the similarity matrix of all one dimensional time series.

5.1 Explicit Relationship

Let $G = \{V, E\}$ be the pre-defined relationship among all one dimensional time series, where V is the node set of time series and E is the set of weight of relationship between two nodes.

The main idea to describe the relationship is using the group effects that all nodes in one group have similar patterns. Here, community detection is adopted to describe those groups in graph from the perspective of complex network theory [3]. One of the most popular methods is modularity based community detection, where the modularity is defined to encourage higher connectivity in group and lower connectivity cross groups:

$$Q = \frac{1}{2m} \sum_{i,j} \left[G_{i,j} - \frac{k_i k_j}{2m} \right] \delta\left(c_i, c_j\right) \tag{3}$$

where $m = \frac{1}{2} \sum_{i,j} G_{i,j}$ indicates all of weights of graph, $k_i = \sum_j G_{i,j}$ represents the degree of node i, $\delta\left(c_i, c_j\right)$ is the function to judge whether i, j are in the same group. There are a number of algorithms have been well studied to optimize the groups by maximizing the modularity. In this

section, the Louvain algorithm is adopted to solve this problem.

5.2 Implicit Relationship

The implicit relationship represents the dependency through similarity among all one-dimensional time series, which needs no additional but only the original time series. A simple approach is that calculate the similarities among those time series though the latent factors. However, one of the greatest challenge is that the similarity cannot be obtained correctly due to the serious bias among them.

Time Index	1	2	3	4	5	6	7	8	9	10
Series 1	3	3	3	3	3	4	4	4	4	5
Series 2	1	1	1	1	1	3	3	3	3	4
Series 3	1	1	1	1	1	1	1	1	1	1

As shown in Table 5.2, Series 1 is a time series which always has higher volume than Series 2, whereas Series 3 is much cold than Series 2. However, the Pearson's correlation coefficient between Series1 and Series 2 (0.9497) is unexpectedly smaller than that between Series 2 and Series 3 (0.9834), although Series 1 and Series 2 have obviously similar patterns.

Here, a decoupled normalization method [1] is introduced to extract the patterns from time series, so that we can explicitly find similar time series in terms of their patterns instead of their bias. This method defines pattern approximated as the likelihood as follows.

$$P(r \text{ from pattern }) = P(x \leq r) - P(x = r)/2 \tag{4}$$

where r indicates the value of time series at certain time index and x indicates all values of this series. $P(x \leq r)$ implements the idea that the how much the likelihood that the value in this time index is higher than others. The extracted pattern of these three single time series is displayed as Table 5.2.

Time Index	1	2	3	4	5	6	7	8	9	10
Series 1	0.25	0.25	0.25	0.25	0.25	0.7	0.7	0.7	0.7	0.95
Series 2	0.25	0.25	0.25	0.25	0.25	0.7	0.7	0.7	0.7	0.95
Series 3	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5

6 Implicit feedback

Implicit feedback is the problem for discreet time series the categorical value cannot represent the real condition explicitly. For example, the traffic data uses four types $\{1, 2, 3, 4\}$ to describe different congestion conditions. Although we know 4 is more congested than 2, but we cannot compare the difference between $1, 3$ and $2, 4$. Generally, the matrix factorization model uses product of latent factors as the prediction for continual variables but fails to prediction for discreet values.

To cope with above shortage improving the accuracy, we uses a classification model to integrate the latent factors and inner dependency to make the final prediction. The classification model can be written as follow:

$$Y_{i,t} = f(P_i, C_i, Q_t|\theta) \tag{5}$$

where C_i is the extracted group specification through the inner dependency among all single time series. To specilization, the classification model used here is Catboost that an unbiased boosting implementation [2].

7 Experiments

In this section, we conduct the empirical analysis to prove the rationality of our framework, Then we compare the core part of our framework the TNMF model with other matrix factorization models in both accuracy and time cost. Finally, we compare the performance of our framework with other cutting-edge time series prediction methods.

7.1 Dataset

To validate our proposed method, a real-word traffic data is used to conduct the experiments. There are real traffic data collected from Xi'an city from July 1, 2019 to July 30, 2019, as well as road attributes and road network topology information. We are required to predict for the road status(i.e. fast, slow or congested) in Aug 1, 2019. This data contains the road status of 15584 road links through 720×30 time points, with only 25.3% nonzero values.

7.2 Experiment setting

To validate for the advantages of TNMF model, three matrix factorization methods are compared:

- SVD: The basic SVD matrix factorization method without considering any temporal dependency.
- TRMF: The temporal regulized matrix factorization method specified by AR model with $q = 50$.
- TNMF: The proposed temporal non-regulized matrix factorization method with $\mathcal{L} = \{0, 1, 2, 720, 720 \times 7\}$.

7.3 Empirical analysis

To show the rationality of our framework, especially show the intuition of the design of TNMF model and why it outperforms other similar approaches, we conducted a series of empirical analysis on the traffic dataset.

First, we draw the average of original time series of all road links. It is apparent from Fig.5 that there is a strong autocorrelation from various granularities, i.e. the autocorrelation from days and autocorrelation from weeks. However, it is almost not possible to investigate this complicated relationship by AR model unless setting super large parameter q, which may leads to unaccepted time consuming or converge issue.

Then, we draw the trained latent factors obtained from TNMF and TRMF. It is clear to see both of them can correctly represent the periodicity, but the TRMF model shows the autocorrelation from long period as the patterns are sightly different in four weeks. That interesting aspect may caused by the large elements in parameter $\mathcal{L}$, which may works like the attention mechanism in deep learning area.

7.4 Performance comparisons

In this section, we intend to compare the performance of the core part TNMF model in our framework with other matrix factorization based model. We use the data of first 30 days as training set, then the last day as test set for evaluation. In Fig 7.1, the proposed TNFM model is shown to significantly outperforms other two models on accuracy with any size of latent factors. In

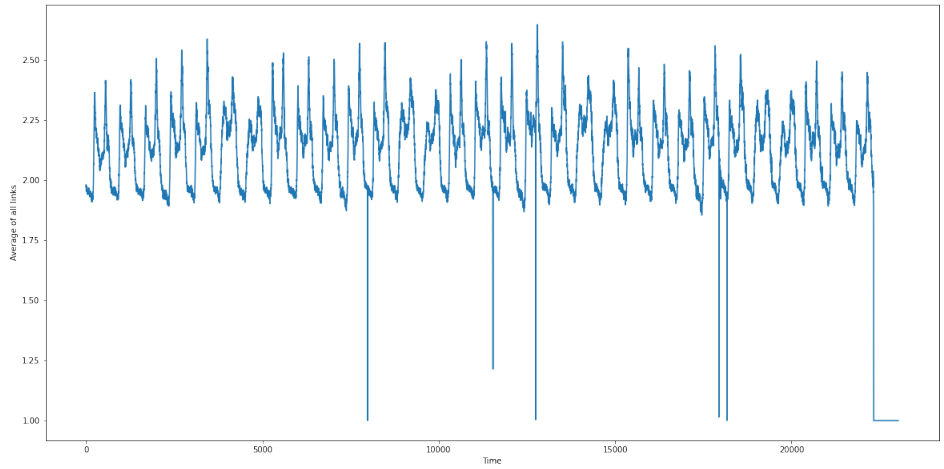


Figure 5: The average original time series of all road links.

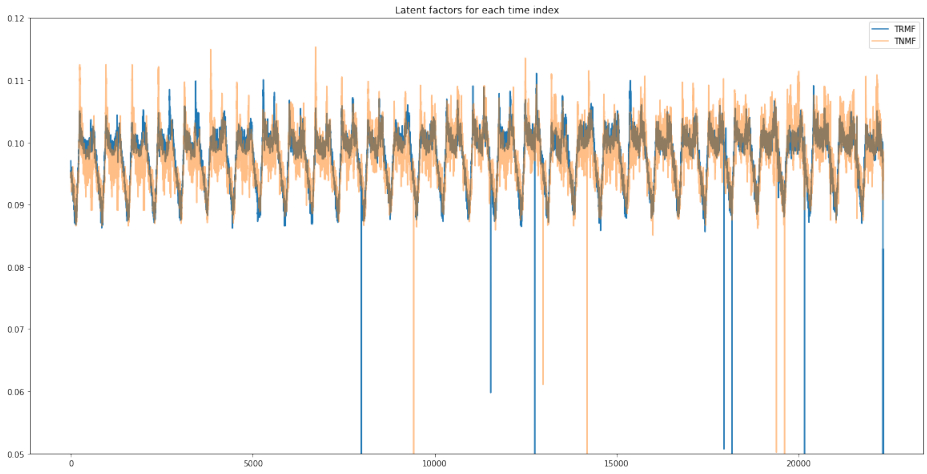


Figure 6: The average latent factors of all dimensions.

Fig 7.2, the proposed TNMF model is slower than other models but the gap between keeps consistent with any all size of K due to the time complexity we analyzed before.

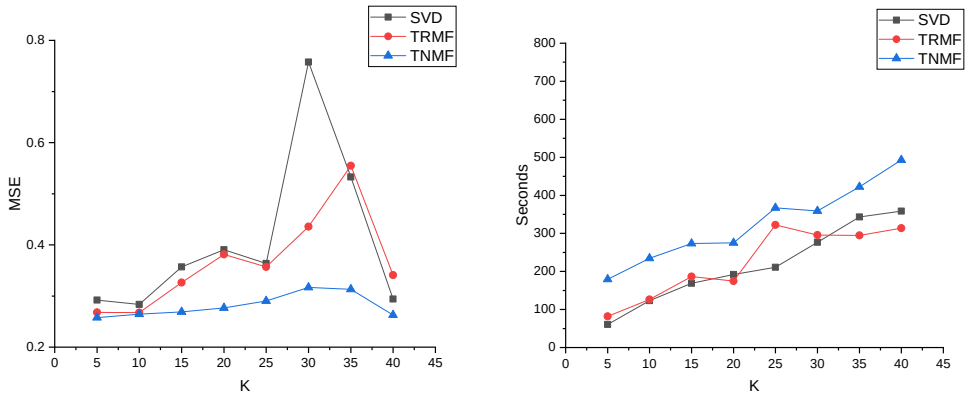


Figure 7: The performance of different MF approaches on test set.

To test the performance of our framework on real applications, we also adopt our framework on the data mining competition at DataFountain[1] platform. Compared with purely using TNMF model, introducing catboost model for implicit feedback can improve around 48% on weighted F1 score, introducing the extracted implicit and explicit inner dependency with catboost can improve around 96% on weighted F1 score. Finally, based on this framework, we achieved the 12-*th* among 2482 teams without intricate parameter turning.

8 Future work

As the due date of this report is tomorrow, there is a lot of work I cannot finish that are expected to be done in the future: 1) The implement of Seq2seq for the prediction of delay latent factors; 2) More comparative experiments on different data set; 3) Making this report more technical.

[1]https://www.datafountain.cn/competitions/466

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